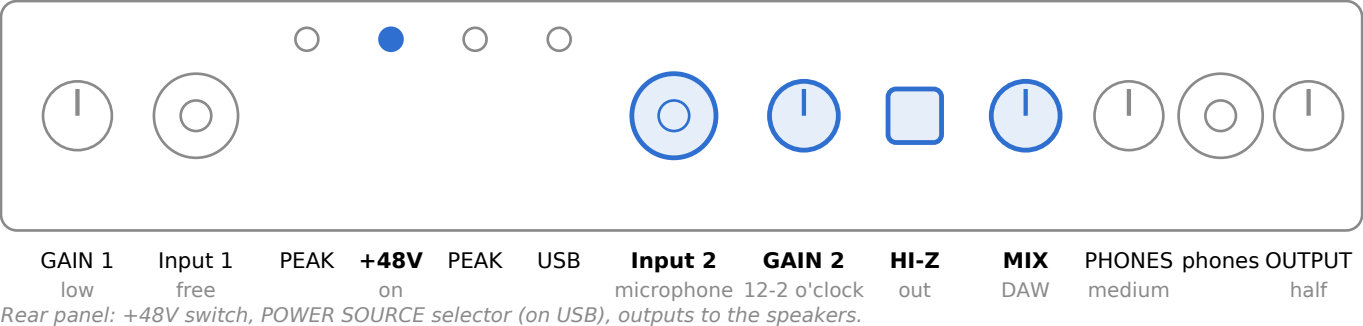


Steinberg UR22mkII on Linux — a short manual

29 September 2026 · Steinberg ST-M01 microphone · Debian 12, PulseAudio

The panel

The five controls to check before every recording are in colour; the others stay as they are.



Control	What it does	Usually
INPUT 1 GAIN	How much input 1 is amplified	Low: nothing is connected
Input 1 (combo)	XLR socket for a microphone or jack for an instrument/line	Free
PEAK	Lights up when the signal of that input is close to clipping	Off, or only on the strongest hits
+48V (LED)	On = phantom power active on both inputs	On, for the ST-M01
USB (LED)	On = interface connected and powered	On
Input 2 (combo)	Same as input 1	ST-M01 microphone
INPUT 2 GAIN	How much input 2 is amplified	2 o'clock guitar, 12 o'clock with voice
INPUT 2 HI-Z	Pressed = input 2 set for an electric guitar on a jack	Out (released) with the microphone
MIX	What you hear: INPUT (left) = inputs only, DAW (right) = computer only, centre = both	DAW to record and play back
PHONES	Volume of the interface headphone socket	To taste
OUTPUT	Volume of the outputs to the speakers	About half
+48V (rear)	Phantom power switch, top right: OFF left, ON right	On with the microphone
POWER SOURCE (rear)	Where the interface takes power from: 5V DC socket or USB	USB : on 5V DC, with no power supply, the interface stays off

The ST-M01 has a blue light: on when it receives phantom power, off when +48V is off.

Connections

What	Where	Settings
ST-M01 microphone (condenser)	Input 2, XLR cable	+48V on, HI-Z out
Acoustic guitar	In front of the microphone, 20-30 cm away	Microphone aimed where the neck meets the body, not at the sound hole; it picks up from the side with the <i>steinberg</i> logo
Electric guitar	Input 2, jack cable	HI-Z pressed, no +48V needed
Speakers	LINE OUTPUT L / R on the rear	Volume with OUTPUT
Headphones	In the speakers' socket	With headphones plugged in, the speakers are mute
PC	USB on the rear	Also powers the interface

Setting the gain

Target: peak around -12 dBFS. The PEAK LED lights only when you are already close to clipping: if it stays off while you play, that is normal.

Measured peak	What it means	What to do
below -30 dBFS	Too low, hiss becomes audible	Turn GAIN up
-18 to -6 dBFS	Right	Nothing
0 dBFS, clipped samples	The sound is cut off and cannot be recovered	Turn GAIN down and try again

Singing close to the microphone raises the signal a lot: on 29 September, with GAIN at 2 o'clock, the recording reached 0 dBFS. Tapping the grille lights PEAK even with little gain.

Monitoring and feedback

MIX	You hear	When
Fully left (INPUT)	The microphone only	To test it, with the speakers muted
Centre	Microphone and computer	To play along with a track, on headphones
Fully right (DAW)	The computer only	To record and to play back

Feedback is the howl of a microphone picking up the speakers. Avoid it with MIX on DAW, or OUTPUT at minimum, or with headphones plugged into the speakers.

Everyday use (music, video)

Spotify, YouTube and the browser play through PulseAudio, which uses the UR22mkII as default output and input. Volume is set with OUTPUT.

Control	Position	Why
MIX	Fully on DAW	You hear the computer only
GAIN 1 and 2	Minimum	No noise from the inputs
+48V	OFF	Only for recording or video calls
Desktop volume	100%, never above	Above 100% PulseAudio distorts
OUTPUT	The actual volume control	9–12 o'clock to listen; clean up to three quarters
Speaker knob	Fixed, set once	Too high and the speakers clip

Setting the speakers, once: desktop volume 100%, OUTPUT at minimum, speakers at half; with a loud track raise OUTPUT to the highest volume you will ever use. If you get there with OUTPUT almost at minimum, or hear clipping, lower the speaker knob and repeat, until the maximum comes with OUTPUT at 12–2 o'clock. Then leave the speaker knob alone.

Reference position (29 September 2026, tested with Hans Zimmer's *Cornfield Chase* and *Time*): desktop volume 100%, **speaker knob at half**. With OUTPUT at **three quarters** the speakers are still clean, with headroom above: already very loud, and for normal listening OUTPUT between 9 and 12 o'clock is enough.

The UR22mkII outputs are at a higher level than many desktop speakers expect: it is the speakers' input that clips, not the interface. If the desk vibrates on bass notes, put some foam under the speakers.

On Linux

Item	Value
Device	hw:1,0 (card 1, Steinberg UR22mkII); check with <code>arecord -l</code>
Format	S32_LE only (32 bit): <code>-f cd</code> fails
Channels	2: left = input 1, right = input 2
Sample rates	44 100 to 192 000 Hz; 44 100 is enough
Software controls	None (amixer <code>-c 1</code> is empty): everything is on the knobs
Playback	<code>aplay file.wav</code> goes through PulseAudio (default output: the UR22). <code>-D plughw:1,0</code> bypasses PulseAudio: only when nothing else is playing
Recording	<code>arecord -D hw:1,0</code> works while PulseAudio plays: the input is free as long as no program uses the microphone

Ready-made commands

Record 30 seconds (MIX on DAW):

```
arecord -D hw:1,0 -f S32_LE -c 2 -r 44100 -d 30 take.wav
```

Record with no time limit, stop with Ctrl+C (with voice: GAIN 2 at 12 o'clock):

```
arecord -D hw:1,0 -f S32_LE -c 2 -r 44100 song.wav
```

Play back (through PulseAudio, even while the browser is playing):

```
aplay song.wav
```

About 21 MB per minute. Count "one, two, three, four" out loud before playing. The file name is relative to the folder you are in.

When something goes wrong

Symptom	Likely cause	Remedy
Silent recording	HI-Z pressed, +48V off, GAIN low, cable	HI-Z out, +48V on, GAIN at 2 o'clock; then try another cable
Microphone not heard on headphones	MIX on DAW	MIX on INPUT, speakers muted
Computer not heard	MIX on INPUT, volumes at zero	MIX on DAW or centre, raise the volume
Loud howl	Feedback	MIX on DAW or OUTPUT at minimum
Bang when pressing HI-Z or +48V	Normal with high volumes	Lower GAIN, PHONES, OUTPUT first
Hiss with GAIN at maximum	Preamp noise	GAIN at 12-2 o'clock
Peak at 0 dBFS	Clipping, often when singing	GAIN 2 towards 12 o'clock
Sample format non available	-f S32_LE missing	Use the ready-made commands
No such file or directory	Wrong path for the current folder	Just the file name, or the full path
Device or resource busy	PulseAudio is already using the interface (browser, Spotify)	Drop -D plughw:1,0: aplay file.wav

Safe order: how to switch the microphone off

The ST-M01 has no switch: it is turned off by switching +48V off. The cable comes off only afterwards.

To start	To finish
1. OUTPUT, PHONES and GAIN 2 at minimum	1. OUTPUT, PHONES and GAIN 2 at minimum
2. Connect the microphone to input 2 (HI-Z out)	2. Switch +48V off: the red LED and the blue light on the microphone go out
3. Switch +48V on, on the rear	3. Only now, if you like, unplug the XLR; it may also stay connected
4. Raise GAIN 2, then OUTPUT and PHONES	

Unplugging the XLR with +48V on makes a loud bang in headphones and speakers and, in the long run, stresses microphone and speakers.